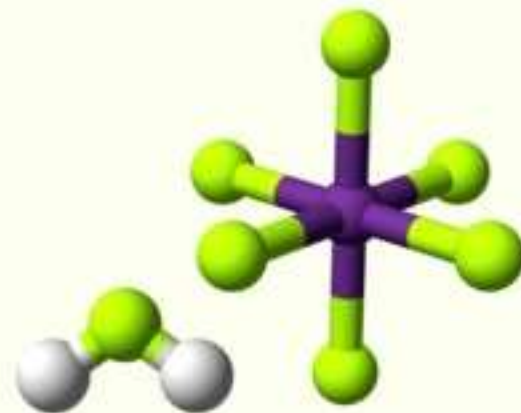


CHEMISTRY

EXTRACTION OF METALS

MODULE - 8



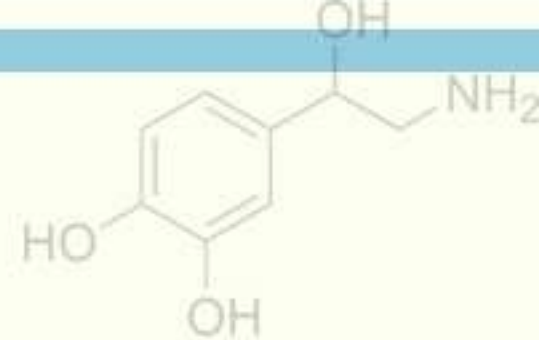


WHERE ARE METALS FOUND ON EARTH?



- Metals occur in the Earth's crust and in sea water; in the form of minerals and ores.
- The Earth's crust is the major source of metals.
- Sea water contains many salts which are compounds of a metal and a non metal; such as sodium chloride, magnesium chloride, etc.





METALS

FREE STATE

COMBINED STATE



Metals from the Earth is extracted from minerals which are either in the combined state or free state.



WHY EVERY MINERAL IS NOT AN ORE?

MINERALS

A mineral is a naturally occurring chemical compound containing some elements.

ORES

The minerals from which a metal can be profitably extracted are called ores.

Profitable over expenses of machines, labour, manpower and time taken





METALLURGY



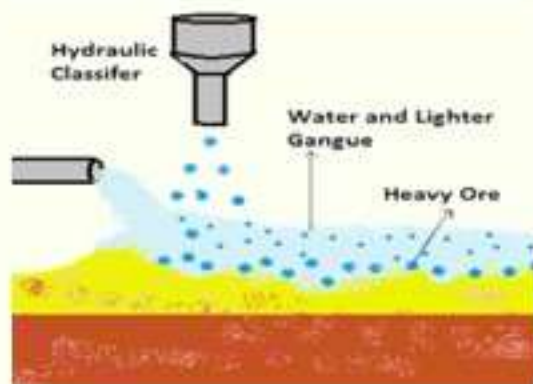
- Metallurgy is a domain of science and engineering of metals for use in products for consumers and manufacturers.
- The production of metals involves the processing of ores to extract the metal they contain, and the mixture of metals, sometimes with other elements, to produce alloys.



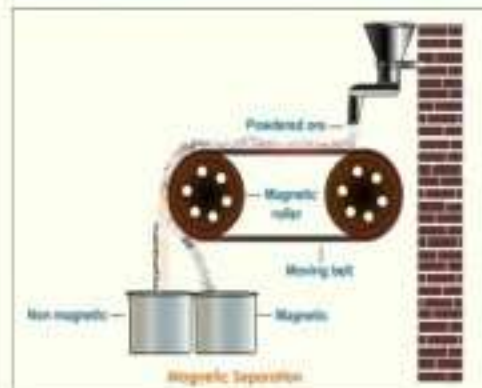
CONCENTRATION OF ORES

Removal of impurities, such as soil, sand, stone, silicates, etc. from mined ore is known as Concentration of Ores.

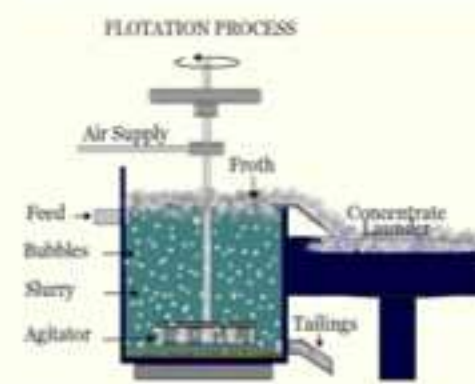
GRAVITY SEPARATION



ELECTROMAGNETIC



FROTH FLOATATION



The unprofitable impurities from an ore is called gangue





THE METAL REACTIVITY SERIES

**HIGHLY
REACTIVE**

Potassium
Sodium
Lithium
Calcium

React with water

**MODERATELY
REACTIVE**

Magnesium
Aluminium
Zinc
Iron
Tin
Lead

React with acids

React with oxygen

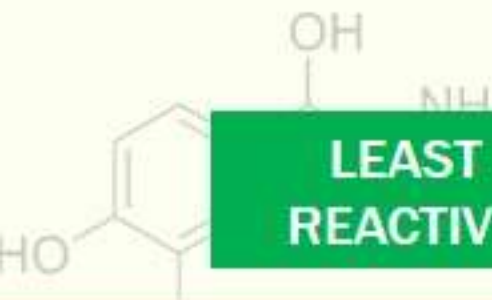
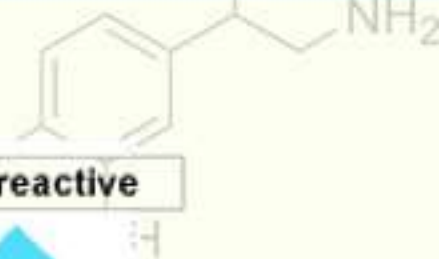
**LEAST
REACTIVE**

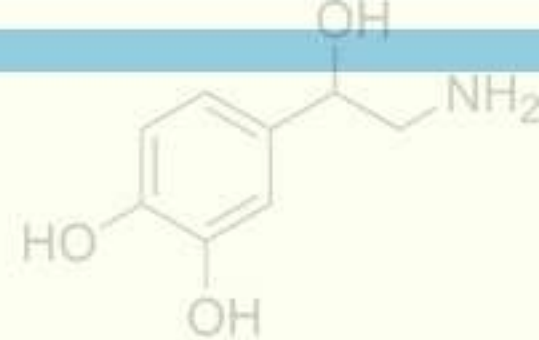
Copper
Mercury
Silver

Gold

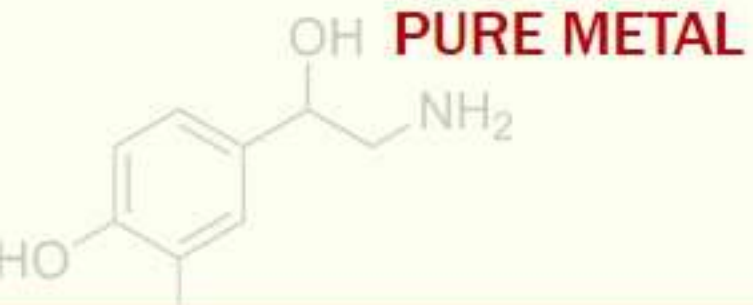
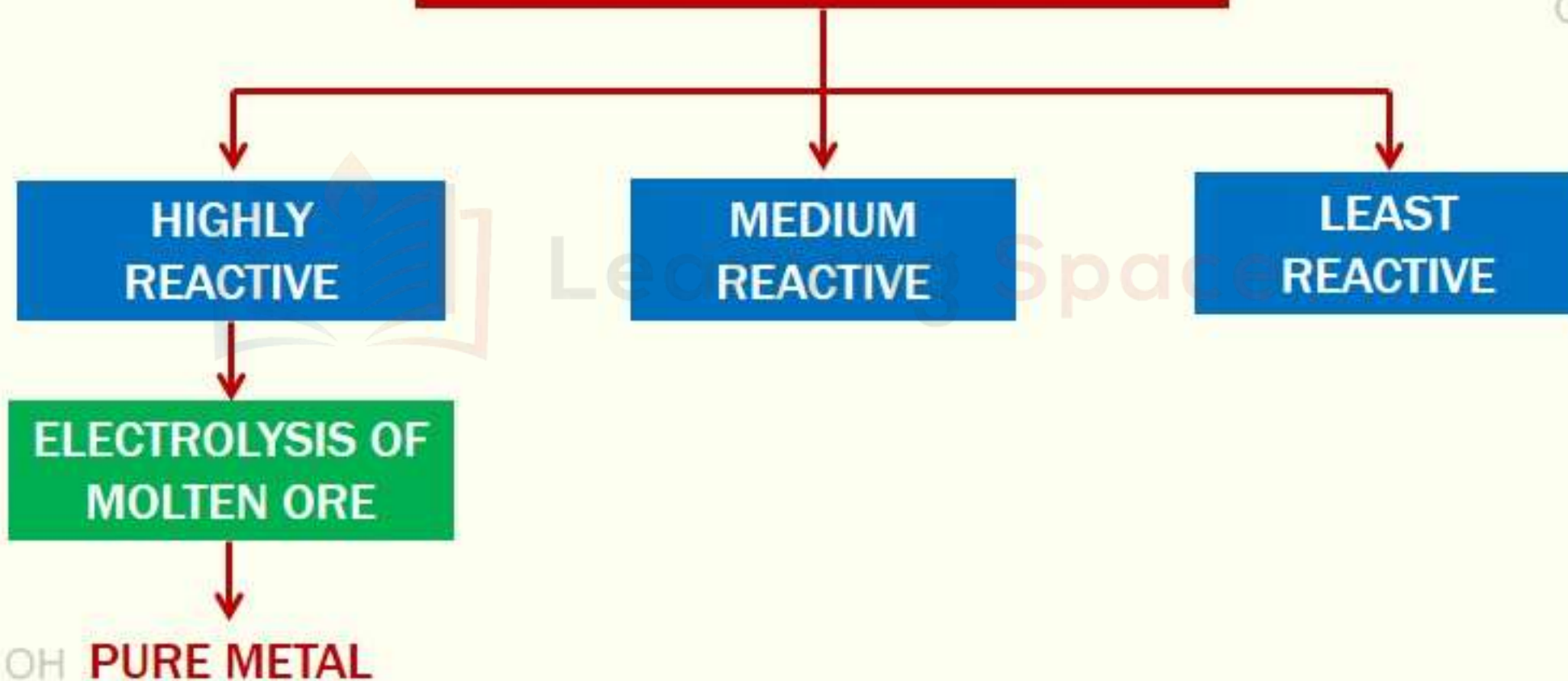
Very reactive

Very unreactive



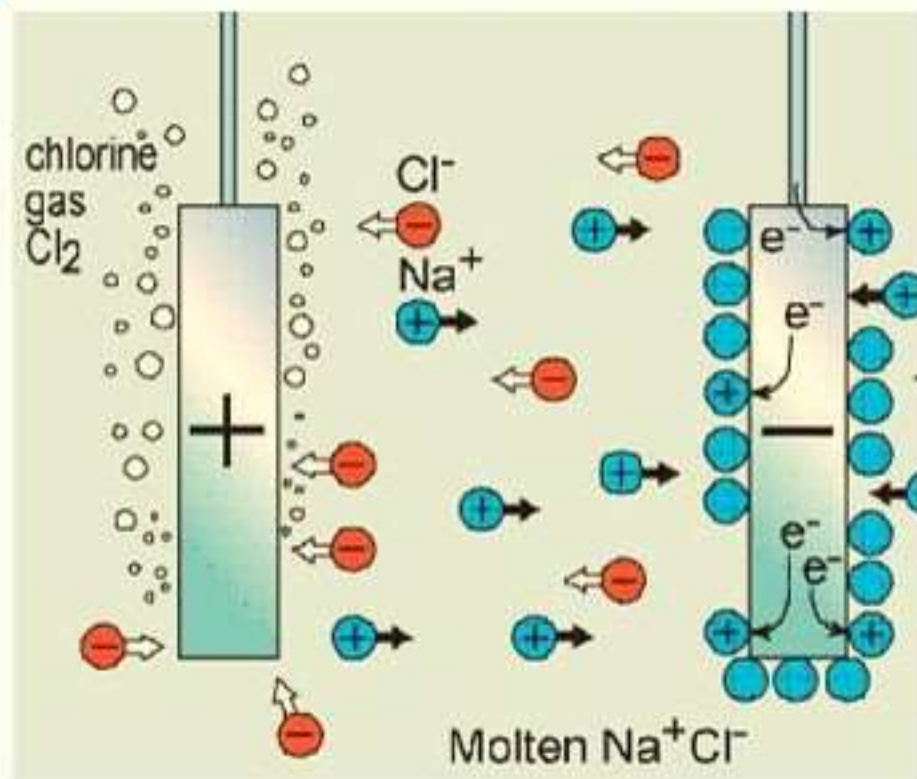


EXTRACTION OF METALS





EXTRACTION OF METALS OF HIGH REACTIVITY



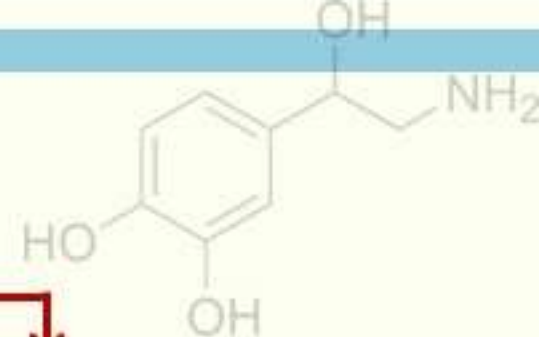
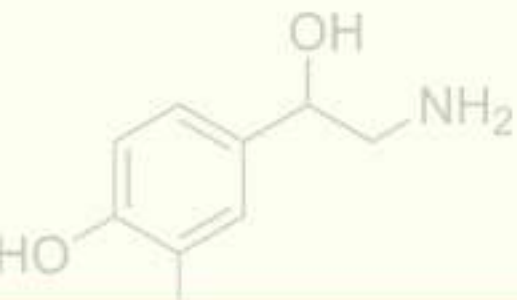
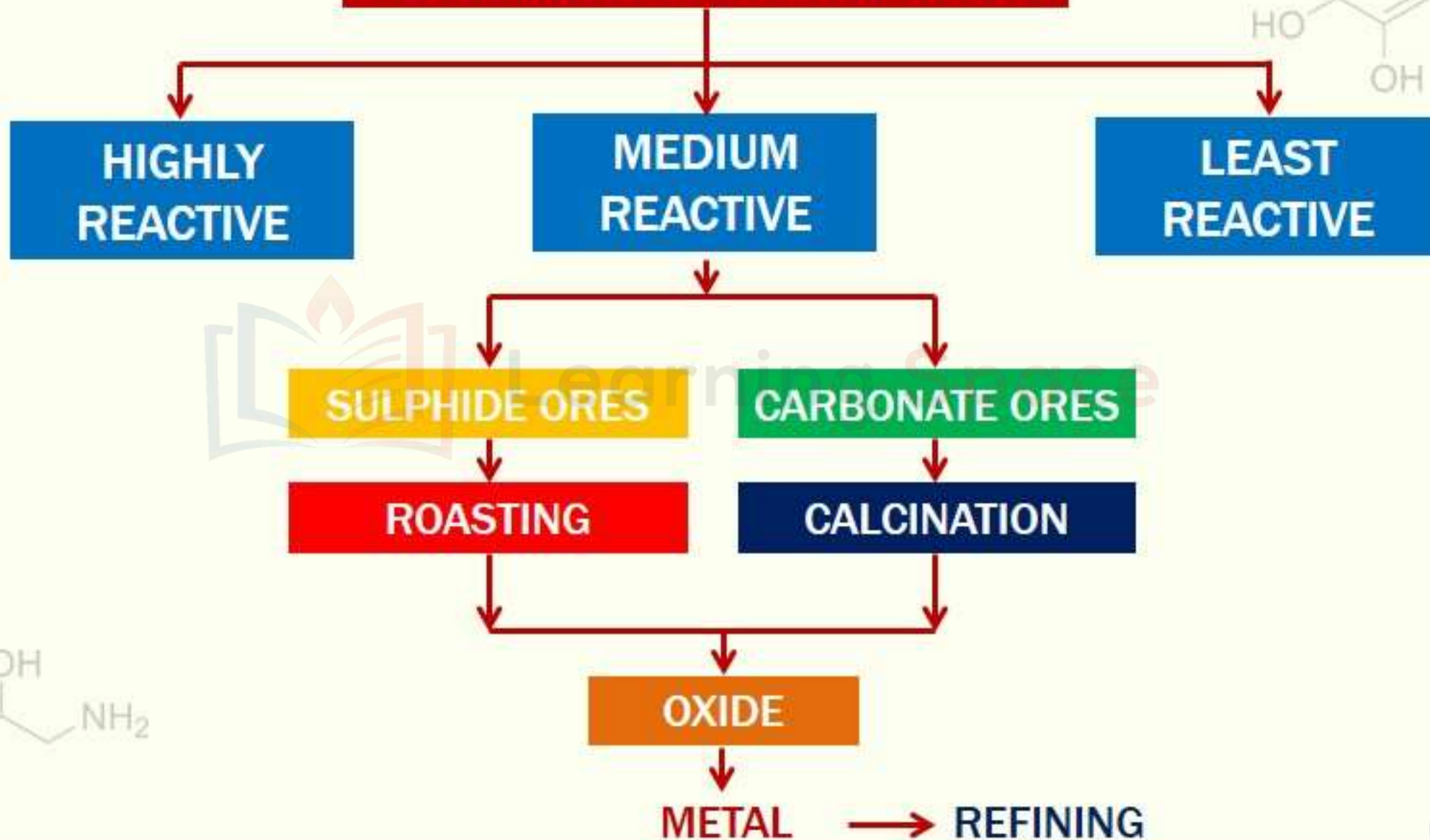
Metals obtained from the process of electrolytic reduction are pure in form.

- Metals of high reactivity, such as sodium, calcium, magnesium, aluminum, etc. are extracted from their ores by **electrolytic reduction**.
- Electrolytic Reduction:** Electric current is passed through the molten state of metal ores. Metal, being positively charged, is deposited over the cathode.

E.g. When electric current is passed through the molten state or solution of sodium chloride, sodium metal deposited over the cathode.



EXTRACTION OF METALS



DIFFERENCE IN THE EXTRACTION PROCESS

ROASTING

- The concentrated ore is heated **in presence of air**
- It is used to convert **sulphide ores into corresponding metal oxide**
- Sulphur dioxide gas is evolved.

CALCINATION

- The concentrated ore is heated **in absence of air**
- It is used to convert **carbonate ores into corresponding metal oxide**
- Carbon dioxide gas is evolved.



ROASTING Vs. CALCINATION

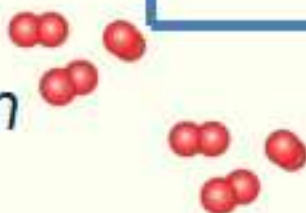


Zinc sulphide
Zinc blende/Sphalerite



Zinc carbonate
Zinc spar/Smithsonite

Oxygen
in air



Zinc oxide





REDUCTION OF THE METAL OXIDE INTO THE METAL BY CARBON AS A REDUCING AGENT



Zinc oxide
 ZnO



Hematite
 Fe_2O_3



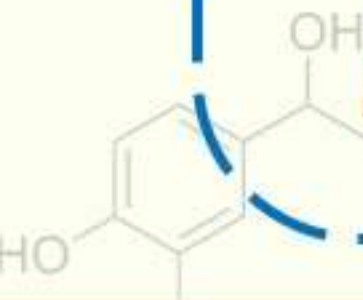
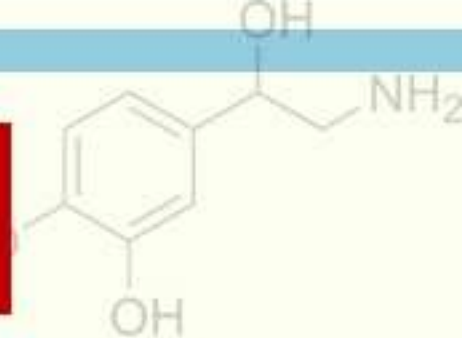
CARBON
Reducing agent



Zinc metal



Iron metal





REDUCTION OF THE METAL OXIDE BY ALUMINUM



- Ferric oxide; when heated with aluminum; is reduced to iron metal.
- In this reaction, lot of heat is produced. This reaction is also known as **Thermite Reaction**.
- Thermite reaction is used in welding of electric conductors, iron joints, etc. such as joints in railway tracks. This is also known as **Thermite Welding (TW)**.
- **Manganese dioxide and copper dioxide** are also reduced by aluminum.





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Lithium
Calcium

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Magnesium
Aluminium
Zinc
Iron
Tin
Lead

React with acids

React with oxygen

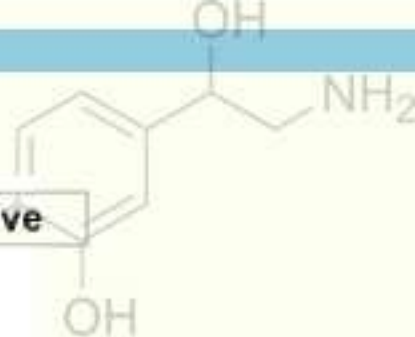
**LEAST
REACTIVE**

Copper
Mercury
Silver

Gold

Very reactive

Very unreactive



EXTRACTION OF METALS

**HIGHLY
REACTIVE**

**MEDIUM
REACTIVE**

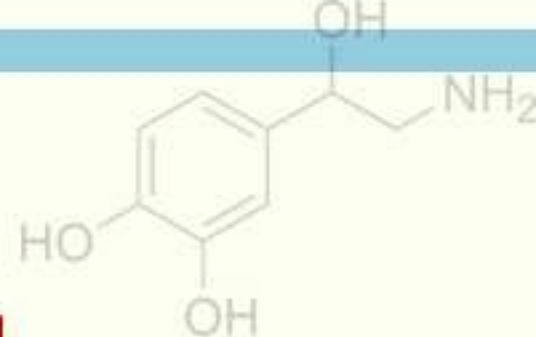
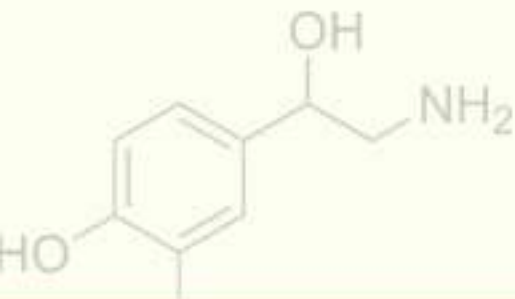
**LEAST
REACTIVE**

SULPHIDE ORES

ROASTING

METAL

REFINING





EXTRACTION OF LOW REACTIVE METALS BY ROASTING

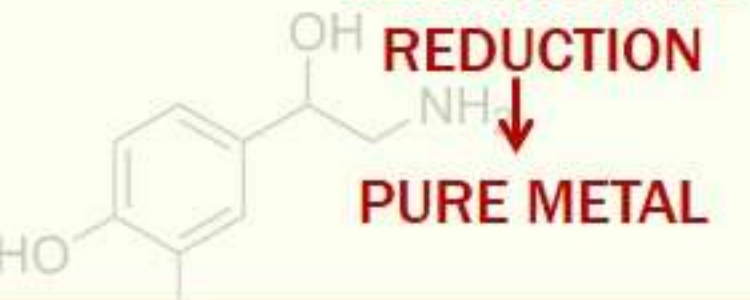
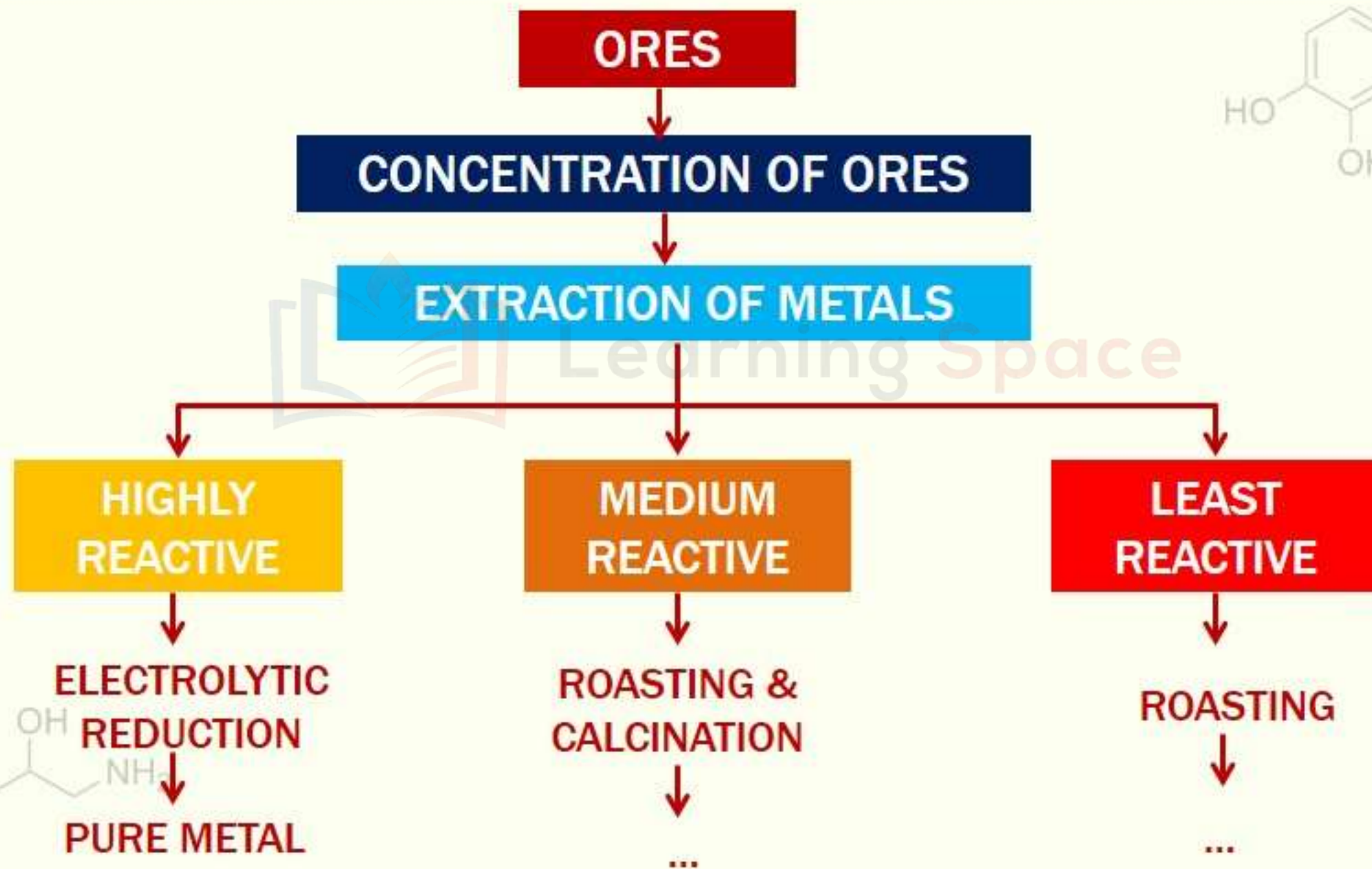
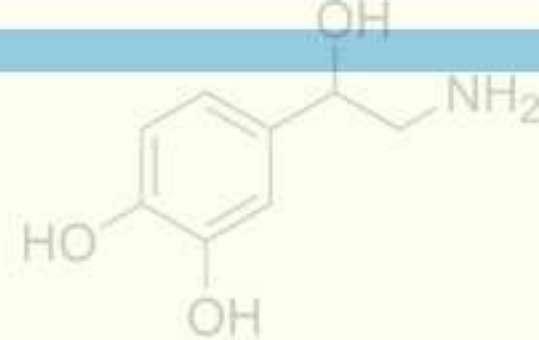
MERCURY SULPHIDE

- **Cinnabar** or HgS is first heated in the air. This turns HgS into HgO .
- Mercury oxide obtained is again heated strongly. This produces mercury metal.

COPPER SULPHIDE

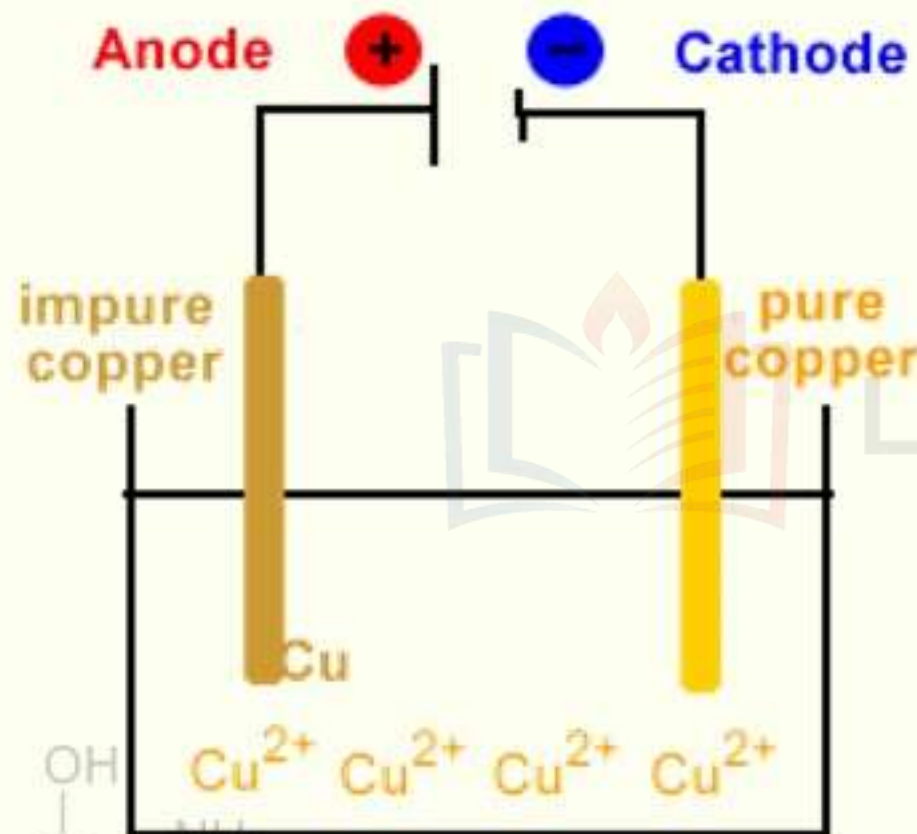
- **Copper glance** or Cu_2S is heated in air turning it into Cu_2O .
- The copper oxide can be heated strongly to obtain pure copper.







ELECTROREFINING OF METALS

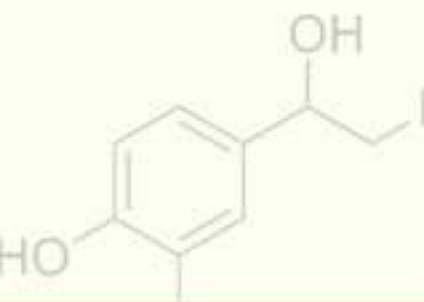
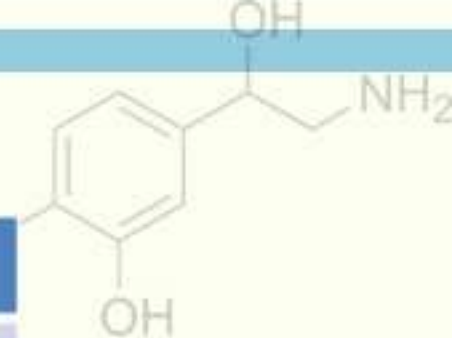


- To obtain 99.9% pure metal, the extracted metals go through a last step of electrorefining.
- In this process, a thick impure metal rod is taken as anode and a pure strip of that metal is taken as the cathode.
- After the process, the metal ions leave the anode and go towards the cathode, thus depositing only pure metal in the cathode.
- The anode leftover is called anode-mud.



METALS AND THEIR RESPECTIVE ORES

METAL	ORES
Sodium	Rock salt
Calcium	Fluorspar, limestone, gypsum
Magnesium	Magnesite, Dolomite
Aluminum	Bauxite, potassium aluminum sulphate
Zinc	Zinc blende, zinc spar, calamine
Iron	Hematite, magnetite
Tin	Cassiterite
Lead	Galena
Copper	Chalcocite, malachite
Mercury	Cinnabar



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